

Tensor Products

Abstract

**Exercise 1.** Less talked about than inner products is the **outer product** of two vectors. If  $u, v \in \mathbb{R}^n$ , their outer product is  $\langle u, v \rangle = uv^T \in M_n(\mathbb{R})$ .

- (a) Show that the map  $\mathbb{R}^n \times \mathbb{R}^n \rightarrow M_n(\mathbb{R})$  given by the outer product is bilinear. That is,
  - i.  $\langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle$ , and similarly for addition in the second component; and,
  - ii. for any scalar  $a$ ,  $\langle au, v \rangle = \langle u, av \rangle = a\langle u, v \rangle$ .
- (b) Suppose  $V$  is a vector space, and suppose  $\phi: \mathbb{R}^n \times \mathbb{R}^n \rightarrow V$  is any bilinear map. Show that there is a linear map  $\tilde{\phi}: M_n(\mathbb{R}) \rightarrow V$  so that  $\phi(u, v) = \tilde{\phi}(\langle u, v \rangle)$ .
- (c) The usual inner product is a bilinear map  $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ . What is the corresponding map  $M_n(\mathbb{R}) \rightarrow \mathbb{R}$ ?

**Exercise 2.** Let  $R$  be an integral domain with fraction field  $K$ , and let  $M$  be a free  $R$  module of rank  $n$ . Prove that any list of  $n + 1$  elements of  $M$  are linearly dependent.

**Exercise 3.** Prove that  $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/3\mathbb{Z})$  is the trivial group. What about  $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/4\mathbb{Z})$ ?

**Exercise 4.** Let  $A$  be a commutative ring, and let  $B$  be an  $A$ -algebra (that is, there is a ring homomorphism  $\phi: A \rightarrow B$  which induces on  $B$  the structure of an  $A$ -module.)

- (a) Show that  $B \otimes_A A[t] \cong B[t]$ .
- (b) Recall that if  $I$  is an ideal of  $A$ , then the extension of  $I$  to  $B$ , denoted  $I^e$ , is the ideal of  $B$  generated by the image  $\phi(I)$ . Prove that  $B/I^e \cong B \otimes_A (A/I)$ .
- (c) Prove that  $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathbb{C} \times \mathbb{C}$  (Hint: rewrite the second  $\mathbb{C}$  in the tensor product as  $\mathbb{R}[x]/(x^2 + 1)$ .)

**Exercise 5.**

- (a) Let  $a, b \in \mathbb{Z}$ . There is a map  $\mathbb{Z} \rightarrow \mathbb{Z}/a\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{b}]$  which sends 1 to  $1 \otimes \frac{1}{b}$ . Find the kernel of this map in terms of  $a$  and  $b$ .
- (b) Now let  $p \in \mathbb{Z}$  be prime, and  $a$  be as before. As above, there is a map  $\mathbb{Z} \rightarrow \mathbb{Z}/a\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}$ . Find the kernel of this map in terms of  $a$  and  $p$ . (Recall: for a prime ideal  $\mathfrak{p} \subset R$ ,  $R_{\mathfrak{p}}$  means we put in all denominators which are *not in*  $\mathfrak{p}$ :  $S = R \setminus \mathfrak{p}$ .)

# Paper

**Exercise 6. (August 2020 Problem 4)** Let  $R$  be a commutative ring.

- (a) Give an example of a free  $R$ -module  $M$  and an injective homomorphism  $M \rightarrow M$  which is not an isomorphism. (For this part,  $R$  can be whatever commutative ring you like.)
- (b) If  $M_1$  and  $M_2$  are free of finite ranks  $r_1$  and  $r_2$ , show that a surjective map  $M_1 \rightarrow M_2$  exists if and only if  $r_1 \geq r_2$ .
- (c) Assume furthermore that  $R$  is an integral domain. Show that an injective map  $M_1 \rightarrow M_2$  exists if and only if  $r_1 \leq r_2$ .

**Exercise 7. (January 2017 Problem 5)** Consider the ring  $R = \mathbb{C}[x]$ .

- (a) Describe all simple  $R$ -modules.
- (b) Give an example of an  $R$ -module that is indecomposable, but not simple. (Recall that a module is *indecomposable* if it cannot be written as a direct sum of non-trivial submodules.)
- (c) Consider  $R$ -modules  $M = R/(x^3 + x^2)$ , and  $N = R/(x^3)$ , and take their tensor product over  $R$ :  $M \otimes_R N$ . It is an  $R$ -module, and in particular a vector space over  $\mathbb{C}$ . What is its dimension over  $\mathbb{C}$ ?
- (d) Let  $M$  be any  $R$ -module such that  $\dim_{\mathbb{C}}(M) < \infty$ , and let  $N = R/(x^3)$  as before. Show that

$$\dim_{\mathbb{C}}(M \otimes_R N) = \dim_{\mathbb{C}} \operatorname{Hom}_R(N, M).$$

**Exercise 8. (August 2017 Problem 1, edited)** Let  $M = \mathbb{C}[x]/(x^2 + x)$  and  $N = \mathbb{C}[x]/(x - 1)$ . Compute the dimension of the following tensor products as vector spaces over  $\mathbb{C}$ :

- (a)  $M \otimes_{\mathbb{C}[x]} N$
- (b)  $M \otimes_{\mathbb{C}} N$
- (c)  $M \otimes_{\mathbb{C}[x^2]} N$ .

**Exercise 9.** Let  $f : M \times N \rightarrow P$  be an  $A$ -module homomorphism. For each  $m \in M$ , define a map  $f_m : N \rightarrow P$  by  $f_m(n) = f(m \otimes n)$ . Show that this defines a isomorphism

$$\operatorname{Hom}_A(M \otimes N, P) \cong \operatorname{Hom}_A(M, \operatorname{Hom}(N, P)).$$

**Exercise 10.** Show that if

$$M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is exact, then so is

$$M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0.$$